9. 7 Power Series

- A special kind of infinite series that may be though of as a *polynomial of infinite degree*.
- Most of the important functions like e^x, $\ln x$, $\sin x$, $\tan^{-1} x$, have useful representations as a power series.

Example: $\dfrac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad , \quad |x| < 1 \ ,$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad , \quad \forall \ x \in \mathrm{R} \ ,$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n \quad , \quad -1 < x \le 1 \ ,$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \quad , \quad \forall \ x \in \mathrm{R}$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \quad , \quad \forall \ x \in \mathrm{R}$$

$$\tan^{-1} x = \sum_{n=1}^{\infty} \frac{(-1)^{2n+1}}{2n+1} x^n \quad , \quad |x| \le 1$$

Definition 1:
A **power series about $x = 0$** is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots .$$

A **power series about $x = a$** is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \cdots + c_n (x-a)^n + \cdots$$

in which the **center** a and the **coefficients** $c_0, c_1, c_2, \cdots, c_n, \cdots$ are constants.

Example 1:

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (x-1)^n = 1 + (x-1) + \frac{1}{2}(x-1)^2 - \frac{1}{6}(x-1)^3 + \cdots + \frac{(-1)^n}{n!}(x-1)^n + \cdots$$

is a power series about $x = 1$ with $c_n = \dfrac{(-1)^n}{n!}$.

Question: For what values of x does a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$ converge?

We may use the **Ratio Test (or Root Test) for absolute convergence** to find x.
When a power series is a **geometric series**, solve $|r| < 1$ to obtain x.

Example 2:

The power series $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots$ is a geometric series with $a = 1$ and $r = x$. The series converges for $|r| = |x| < 1$.

Therefore, $\boxed{\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad , \quad |x| < 1}$.

We say that $\sum_{n=0}^{\infty} x^n$ is a power series representation of the function $f(x) = \frac{1}{1-x}$, $|x| < 1$.

When $|x| \geq 1$, $\sum_{n=0}^{\infty} x^n$ diverges, so it can't represent $f(x) = \frac{1}{1-x}$ at these points.

For $|x| < 1$, we can use $\frac{1}{1-x}$ to approximate the k-th partial sum, that is

$$\sum_{n=0}^{k} x^n = 1 + x + x^2 + \cdots + x^k \approx \frac{1}{1-x}.$$

For values of x near 0, we need take only a few terms of the series to get good approximation, but as we move towards $x = 1$ or $x = -1$, we must take more terms (see Figure 1).

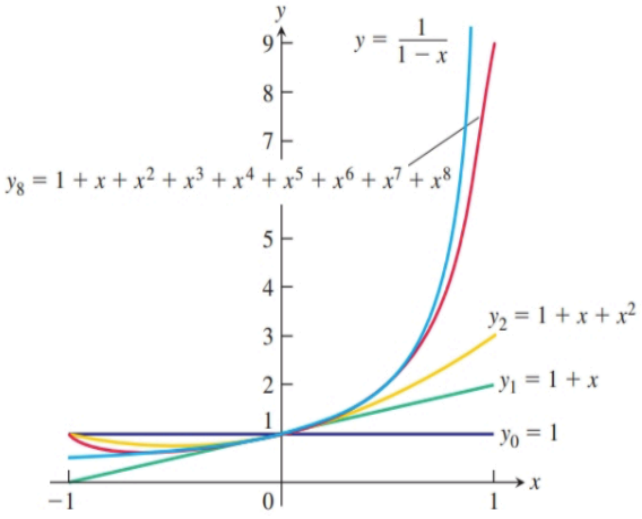


Figure 1

Example 3:

The power series $1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 - \frac{1}{8}(x-2)^3 + \cdots + \left(\frac{-1}{2}\right)^n (x-2)^n + \cdots$ is a geometric series with $a = 2$ and $r = -\frac{x-2}{2}$. The series converges for $|r| = \left|-\frac{x-2}{2}\right| < 1$ or $0 < x < 4$.

The sum is $\frac{a}{1-r} = \frac{1}{1 + \frac{x-2}{2}} = \frac{2}{x}$, so

$$\frac{2}{x} = 1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 - \frac{1}{8}(x-2)^3 + \cdots + \left(\frac{-1}{2}\right)^n (x-2)^n + \cdots, \quad 0 < x < 4.$$

Theorem 1: The Convergence Theorem for Power Series

If the power series $\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots$ converges at $x = c \neq 0$, then it converges absolutely for all x with $|x| < |c|$.

If the series diverges at $x = d$, then it diverges for all x with $|x| > |d|$.

Remark: ***Let*** $\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a)^n + c_2 (x-a)^2 + \cdots + c_n (x-a)^n + \cdots$.

When $x = a$, $\sum_{n=0}^{\infty} c_n (x-a)^n = c_0$. So, $\sum_{n=0}^{\infty} c_n (x-a)^n$ **certainly converges (to c_0) at $x = a$** (its center of convergence).

Suppose R is a positive real number, called the ***radius of convergence*** of a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$, and the interval of radius R centered at $x = a$ is called **the interval of convergence**.

Corollary to Theorem 1:

The convergence of the series

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \cdots + c_n (x-a)^n + \cdots$$

is described by one of the following three cases:

1. There is a positive number R such that the series diverges for x with $|x - a| > R$ but converges absolutely for x with $x - a| < R$. The series may or may not converge at either of the endpoints $x = a - R$ and $x = a + R$.
2. The series converges absolutely for every x ($R = \infty$).
3. The series converges at x = a and diverges elsewhere ($R = 0$).

By **Corollary to Theorem 1**, $\sum_{n=0}^{\infty} c_n (x-a)^n$ behaves in one of three possible ways:

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

converges only at the $x = c$, and diverges elsewhere. ($R = 0$)	converges on some interval of radius R. It converges absolutely for $\|x - a\| < R$ and diverges for $\|x - a\| < R$. The power series may or may not converge at either of the end points $x = a - R$ and $x = a + R$.	converges absolutely for all real x. ($R = \infty$)

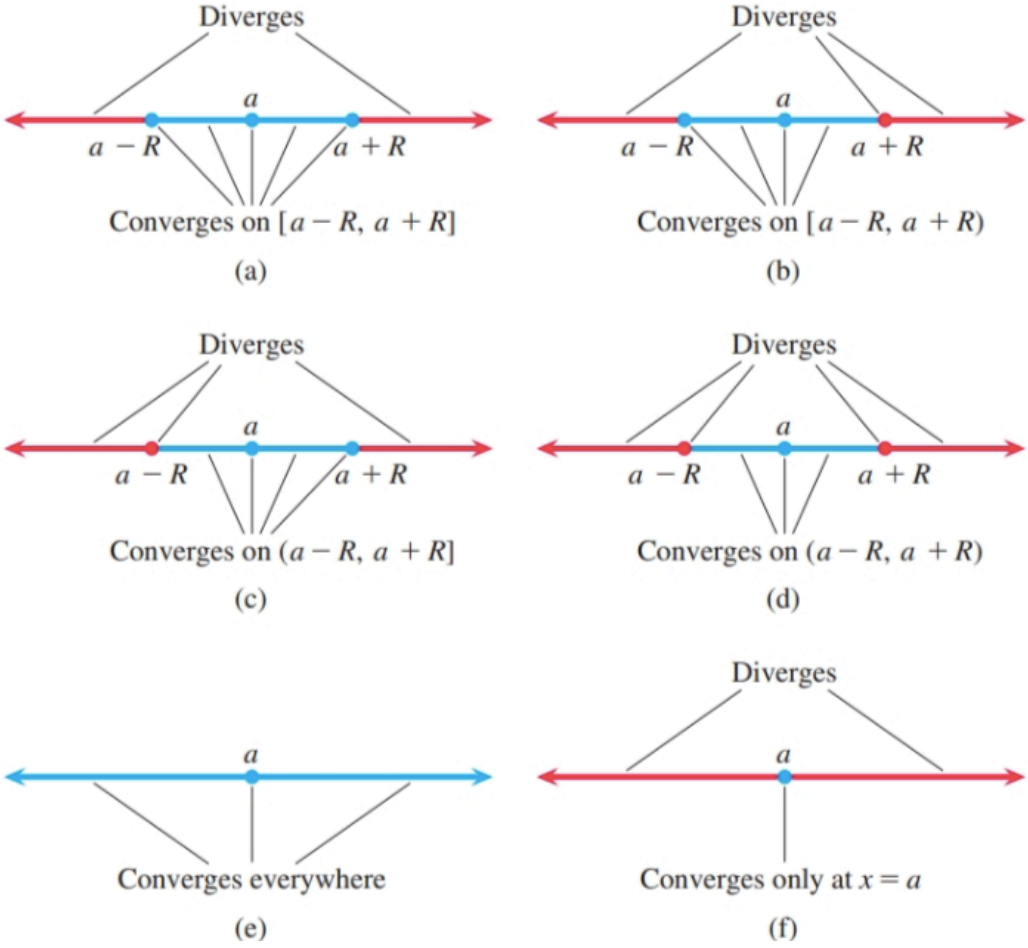


Figure 2 The six possibilities for an interval of convergence.

How to Test a Power Series for Convergence

1. Use the Ratio Test (or Root Test) to find the largest open interval where the series converges absolutely. Ordinarily, this is an open interval

$$|x - a| < R \qquad \text{or} \qquad a - R < x < a + R.$$

2. If the interval of absolute convergence is finite, test for convergence or divergence at each endpoint, as in Examples 4a and b. Use a Comparison Test, the Integral Test, or the Alternating Series Test.

3. If the interval of absolute convergence is $a - R < x < a + R$, the series diverges for $|x - a| > R$ (it does not even converge conditionally) because the nth term does not approach zero for those values of x.

<u>Testing power series for convergence</u> (Use Ratio Test for Absolute Convergence)

Theorem 2: The Ratio Test for Absolute Convergence

Let $\sum a_n$ be a series with ***nonzero*** terms and suppose $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \rho$.

(1) If $\rho < 1$, $\sum|a_n|$ **converges and hence** $\sum a_n$ **converges (absolutely).**

(2) If $\rho > 1$ or ∞, $\sum a_n$ **diverges.** (3) If $\rho = 1$, the test **inconclusive**.

Remark:

If $\sum_{n=0}^{\infty} c_n (x-a)^n$ converges absolutely when $|x-a| < R$ and diverges when $|x-a| > R$,

then R is called the **radius of convergence** of the power series.

$R = 0$, if it converges at $x = a$ only; $R = \infty$, if it converges for all real x.

Example 4:

For what values of x do the following power series converge?

(a) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$

(b) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$

(c) $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} - \cdots$

(d) $\sum_{n=0}^{\infty} n! x^n = 1 + x + 2!x^2 + 3!x^3 + \cdots$

Solutions:

(a) Use the Ratio Test for Absolute Convergence.

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(-1)^n x^{n+1}}{n+1} \left(\frac{n}{(-1)^{n-1} x^n} \right) \right| = |x| \lim_{n\to\infty} \frac{n}{n+1} = |x| .$$

By the ratio test for absolute convergence, the series converges absolutely for $|x| < 1$, and diverges for $|x| > 1$.

When $x = 1$, the series is $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is the alternating harmonic series which converges conditionally.

When $x = -1$, the series is $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}(-1)^n}{n} = -\sum_{n=1}^{\infty} \frac{1}{n}$ is the negative of the harmonic series. $\therefore$ The series diverges.

Therefore, $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$ converges for $-1 < x \le 1$ and diverges elsewhere.

Remark: The radius of convergence R is 1, the interval of convergence is $(-1, 1]$ and The interval of absolute convergence is $(-1, 1)$. The series converges conditionally for $x = 1$.

(b) $\lim\limits_{n\to\infty}\left|\dfrac{a_{n+1}}{a_n}\right| = \lim\limits_{n\to\infty}\left|\dfrac{(-1)^n x^{2(n+1)-1}}{2(n+1)-1}\left(\dfrac{2n-1}{(-1)^{n-1}x^{2n-1}}\right)\right| = \left|x^2\right| \lim\limits_{n\to\infty}\dfrac{2n-1}{2n+1} = x^2 \lim\limits_{n\to\infty}\dfrac{2-\frac{1}{n}}{2+\frac{1}{n}} = x^2$.

$\lim\limits_{n\to\infty}\left|\dfrac{a_{n+1}}{a_n}\right| < 1 \Rightarrow x^2 < 1 \Rightarrow -1 < x < 1.$

By the ratio test for absolute convergence, the series converges absolutely for $-1 < x < 1$, and diverges for $x > 1$ or $x < -1$.

When $x = 1$, the series is $\sum\limits_{n=1}^{\infty}\dfrac{(-1)^{n-1}}{2n-1}$, is an alternating series.

Let $u_n = \dfrac{1}{2n-1}$, $n \geq 1$. Then

(1) $u_n > 0 \quad \forall n \geq 1$;

(2) $u_n = \dfrac{1}{2n-1} > \dfrac{1}{2(n+1)-1} = u_{n+1} \quad \forall n \geq 1$;

(3) $\lim\limits_{n\to\infty} u_n = \lim\limits_{n\to\infty}\dfrac{1}{2n-1} = 0$.

By the alternating series test, the series converges.

When $x = -1$, the series is $\sum\limits_{n=1}^{\infty}\dfrac{(-1)^{3n-2}}{2n-1} = \sum\limits_{n=1}^{\infty}\dfrac{(-1)^n}{2n-1}$ is a convergent alternating series.

Therefore, $\sum\limits_{n=1}^{\infty}\dfrac{(-1)^{n-1}x^{2n-1}}{2n-1}$ converges for $-1 \leq x \leq 1$ and diverges elsewhere.

Remark: The radius of convergence is 1, the interval of convergence is [−1, 1] and the interval of absolute convergence is (−1, 1). The series converges conditionally for $x = \pm 1$.

(c) $\lim\limits_{n\to\infty}\left|\dfrac{a_{n+1}}{a_n}\right| = \lim\limits_{n\to\infty}\left|\dfrac{x^{n+1}}{(n+1)!}\left(\dfrac{n!}{x^n}\right)\right| = \left|x\right| \lim\limits_{n\to\infty}\dfrac{1}{n+1} = 0 < 1 \quad \forall\ x$.

By the ratio test for absolute convergence, the series converges absolutely for all real x. The radius of convergence is ∞ and the interval of convergence is (−∞, ∞).

(d) $\lim\limits_{n\to\infty}\left|\dfrac{a_{n+1}}{a_n}\right| = \lim\limits_{n\to\infty}\left|\dfrac{(n+1)!\,x^{n+1}}{n!x^n}\right| = \left|x\right| \lim\limits_{n\to\infty}(n+1) = \infty$ except $x = 0$.

When $x = 0$, $\sum\limits_{n=1}^{\infty} n!\,x^n = 1+0+0+0+\cdots = 1$, the series converges.

Therefore, the series diverges for all nonzero real x.

The radius of convergence is 0.

Example 5:

For the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{\sqrt{n^2+5}}$, find the interval of convergence and the value of x for which the series converges conditionally.

Solution:

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(x-2)^{n+1}}{\sqrt{(n+1)^2+5}}\left(\frac{\sqrt{n^2+5}}{(x-2)^n}\right)\right| = |x-2|\lim_{n\to\infty}\sqrt{\frac{n^2+5}{n^2+2n+6}}$$

$$= |x-2|\lim_{n\to\infty}\sqrt{\frac{1+\frac{5}{n^2}}{1+\frac{2}{n}+\frac{6}{n^2}}} = |x-2|.$$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1 \Rightarrow |x-2|<1 \Rightarrow -1<x-2<1 \Rightarrow 1<x<3.$$

By the ratio test for absolute convergence, the series converges absolutely for $1 < x < 3$ and diverges for $x < 1$ or $x > 3$.

When $x = 3$, the series is $\sum_{n=0}^{\infty} \frac{1}{\sqrt{n^2+5}}$. Take $a_n = \frac{1}{\sqrt{n^2+5}}$, $b_n = \frac{1}{n}$.

$\sum_{n=1}^{\infty} b_n$ is a divergent series and

$$\lim_{n\to\infty}\frac{a_n}{b_n} = \lim_{n\to\infty}\frac{n}{\sqrt{n^2+5}} = \lim_{n\to\infty}\frac{n}{n\sqrt{1+(5/n^2)}} = \lim_{n\to\infty}\frac{1}{\sqrt{1+(5/n^2)}} = 1 > 0.$$

By the limit comparison test, the series diverges.

When $x = 1$, the series is $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+5}}$, an alternating series.

Let $u_n = \frac{1}{\sqrt{n^2+5}}$, $n \geq 1$. Then

(1) $u_n > 0 \quad \forall n \geq 1$;

(2) $u_n = \frac{1}{\sqrt{n^2+5}} > \frac{1}{\sqrt{(n+1)^2+5}} = u_{n+1} \quad \forall n \geq 1;$

(3) $\lim_{n\to\infty} u_n = \lim_{n\to\infty}\frac{1}{\sqrt{n^2+5}} = 0.$

By the alternating series test, the series converges.

$\sum_{n=1}^{\infty}\left|\frac{(-1)^n}{\sqrt{n^2+5}}\right|$ diverges $\Rightarrow \sum_{n=1}^{\infty}\frac{(-1)^n}{\sqrt{n^2+5}}$ converges conditionally.

$\therefore$ The interval of convergence is [1, 3) and the series converges conditionally at $x = 1$.

Theorem 2: Series Multiplication for Power Series

If $A(x)=\sum_{n=0}^{\infty}a_n x^n$ and $B(x)=\sum_{n=0}^{\infty}b_n x^n$ converges absolutely for $|x|<R$, and

$$c_n = a_0b_n + a_1b_{n-1} + a_2b_{n-2}+\cdots+ a_nb_0 = \sum_{k=0}^{\infty}a_k b_{n-k},$$

then $\sum_{n=0}^{\infty}c_n x^n$ converges to $A(x)B(x)$ for $|x|<R$:

$$\left(\sum_{n=0}^{\infty}a_n x^n\right).\left(\sum_{n=0}^{\infty}b_n x^n\right)=\sum_{n=0}^{\infty}c_n x^n.$$

Example 6:

$$\left(\sum_{n=0}^{\infty}x^n\right).\left(\sum_{n=0}^{\infty}(-1)^n\frac{x^{n+1}}{n+1}\right)=\left(1+x+x^2+\cdots\right).\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots\right)$$

$$=\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots\right)+\left(x^2-\frac{x^3}{2}+\frac{x^4}{3}-\cdots\right)+\left(x^3-\frac{x^4}{2}+\frac{x^5}{3}-\cdots\right)+\cdots$$

$$=x+\frac{x^2}{2}+\frac{5x^3}{3}-\frac{x^4}{6}\cdots.$$

We can also substitute function $f(x)$ for x in a convergent power series.

Theorem 3: Series Multiplication for Power Series

If $A(x)=\sum_{n=0}^{\infty}a_n x^n$ converges absolutely for $|x|<R$, then $\sum_{n=0}^{\infty}a_n(f(x))^n$ converges absolutely for any continuous function f on $|f(x)|<R$.

Example 7:

$\frac{1}{1-x}=\sum_{n=1}^{\infty}x^n$ converges absolutely for $|x|<1$, it follows from Theorem 3 that $\frac{1}{1-4x^2}=\sum_{n=1}^{\infty}(4x^2)^n$ converges absolutely for $|4x^2|<1$ or $|x|<½$.

Theorem 4: Term-by-Term Differentiation

If $\sum_{n=0}^{\infty} c_n (x-a)^n$ has radius of convergence $R > 0$, it defines a function

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n \quad \text{on the interval } a - R < x < a + R.$$

This function f has derivatives of all orders inside the interval, and we obtain the derivatives by differentiating the original series term by term:

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1},$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1) c_n (x-a)^{n-2},$$

and so on. Each of these derived series converges at every point of the interval $a - R < x < a + R$.

Example 8:

Find series for $f'(x)$ and $f''(x)$ if

$$f(x) = \frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots = \sum_{n=1}^{\infty} x^n, \qquad -1 < x < 1.$$

Solution: We differentiate the power series on the right term by term:

$$f'(x) = \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \cdots + nx^{n-1} + \cdots = \sum_{n=1}^{\infty} n x^{n-1}, \qquad -1 < x < 1;$$

$$f''(x) = \frac{2}{(1-x)^3} = 2 + 6x + 12x^2 + \cdots + n(n-1)x^{n-2} + \cdots = \sum_{n=1}^{\infty} n(n-1) x^{n-2}, \qquad -1 < x < 1.$$

Caution:

Term-by-term differentiation might not work for other kinds of series. For example, the trigonometric series $\sum_{n=1}^{\infty} \frac{\sin(n!x)}{n^2}$ converges for all x.But if we differentiate term by term we get the series $\sum_{n=1}^{\infty} \frac{n!\cos(n!x)}{n^2}$ which diverges for all x. This is not a power series since it is not a sum of positive integer powers of x.

It is also true that a power series can be integrated term by term throughout its interval of convergence.

Theorem 5: Term-by-Term Integration Theorem

Suppose that $f(x)=\sum_{n=0}^{\infty} c_n (x-a)^n$ converges for $a-R < x < a+R$ $(R>0)$.

Then $\sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$ converges for $a-R < x < a+R$ and

$$\int f(x)\,dx = \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1} + C \quad \text{for} \quad a-R < x < a+R\,.$$

Example 9:

Identify the function $f(x)=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{2n-1}}{2n-1} = x-\frac{x^3}{3}+\frac{x^5}{5}-\cdots$, $-1 \le x \le 1$.

Solution: We differentiate the original power series and get

$$f'(x) = 1-x^2+x^4-x^6+\cdots, \quad -1<x<1.$$

This is a geometric series with first term 1 and ratio $-x^2$, so

$$f'(x)=\frac{1}{1-(-x^2)}=\frac{1}{1+x^2}.$$

We can now integrate $f'(x)=\frac{1}{1+x^2}$ to get $\int f'(x)\,dx = \int \frac{dx}{1+x^2} = \tan^{-1}x + C.$

This series fro $f(x)$ is zero when $x = 0$, so $C = 0$. Hence

$$f(x) = x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\cdots = \tan^{-1}x, \quad -1<x<1.$$

It can be shown that the series also converges to $\tan^{-1}x$ at the endpoints $x = \pm 1$, but we omit the proof.

Example 10:

The series $\frac{1}{1+t} = 1-t+t^2-t^3+\cdots$ converges on the open interval $-1 < x < 1$. Therefore,

$$\ln(1+x) = \int_0^x \frac{dt}{1+t} = t-\frac{t^2}{2}+\frac{t^3}{3}-\frac{t^4}{4}+\cdots\Big]_0^x = x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots \quad \text{or}$$

$$\ln(1+x) = \sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^n}{n}, \quad -1<x<1.$$

It can also be shown that the series converges at $x = 1$ to the number ln 2, but that was not guaranteed by the theorem.

Remarks:

When a power series is differentiated or integrated term by term, the **radius of convergence** remains the same. How about the **interval of convergence**?

The interval of convergence of the new series same as the original series **EXCEPT possibly at one or both end points.**

Example 11:

For what values of x does the series

$$1-\frac{1}{2}(x-3)+\frac{1}{4}(x-3)^2+\cdots+(x-3)^n+\cdots$$

converge? What is its sum? What series do you get if you differentiate the given series term by term? For what values of x does the new series converge? What is its sum?

Solution:

$$\sum_{n=0}^{\infty}\left(-\frac{1}{2}\right)^n(x-3)^n=\sum_{n=0}^{\infty}(-1)^n\left(\frac{x-3}{2}\right)^n.$$

The series is a geometric series with $a=1,\ r=-\frac{x-3}{2}$.

So, $\sum_{n=0}^{\infty}\left(-\frac{1}{2}\right)^n(x-3)^n=\frac{1}{1+\frac{x-3}{2}}=\frac{2}{x-1},\quad 1<x<5.$

Let $f(x)=\frac{2}{x-1}$, $1<x<5$.

Then $f(x)=1-\frac{1}{2}(x-3)+\frac{1}{4}(x-3)^2-\frac{1}{8}(x-3)^3+\cdots+\left(-\frac{1}{2}\right)^n(x-3)^n+\cdots$.

$$f'(x)=-\frac{1}{2}+\frac{1}{2}(x-3)-\frac{3}{8}(x-3)^2+\cdots+\left(-\frac{1}{2}\right)^n n(x-3)^{n-1}+\cdots=\sum_{n=1}^{\infty}\left(-\frac{1}{2}\right)^n n(x-3)^{n-1},\ 1<x<5.$$

When $x=1$, the series is $\sum_{n=1}^{\infty}\left(-\frac{1}{2}\right)^n n(-2)^{n-1}=\sum_{n=1}^{\infty}-\frac{n}{2}$. $\lim_{n\to\infty}-\frac{n}{2}=-\infty \Rightarrow$ the series diverges.

When $x=5$, the series is $\sum_{n=1}^{\infty}\left(-\frac{1}{2}\right)^n n(2)^{n-1}=\sum_{n=1}^{\infty}\frac{(-1)^n n}{2}$. $\lim_{n\to\infty}\frac{(-1)^n n}{2}$ does not exist $\Rightarrow$ the series diverges.

Therefore, the new series converges for $1<x<5$.

$$\therefore\qquad \sum_{n=1}^{\infty}\left(-\frac{1}{2}\right)^n n(x-3)^{n-1}=-\frac{2}{(x-1)^2},\ 1<x<5.$$